

DOSSO, Niger

WMO: 610530

Lat: 13.049N

Lon: 3.223E

Elev: 738

StdP: 14.31

Time Zone: 1.00 (E01)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.9	63.8	17.7	13.8	85.1	19.0	14.7	83.6	14.1	75.7	12.9	77.1	2.8	N/A	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.7	107.4	71.4	106.0	71.1	104.4	71.0	80.8	93.7	80.0	92.4	79.4	91.5	5.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	150.2	84.5	77.4	146.3	83.9	76.8	143.5	83.6	45.0	93.7	44.2	92.6	43.6	91.4	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
12.0	10.3	9.3	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	44.4	82.9	1.2	2.3	43.5	84.5	42.8	85.8	42.1	87.1	41.3	88.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.4	77.1	83.3	90.0	94.4	93.4	88.7	83.9	81.4	83.6	86.9	83.7	77.9
	DBStd	6.26	3.92	4.34	3.24	2.88	3.81	4.05	3.27	2.99	3.03	2.51	2.20	3.41
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12903	840	934	1239	1331	1345	1161	1050	974	1009	1144	1012	864
	CDD65	7428	375	514	774	881	880	711	585	509	559	679	562	399
	CDH74	99294	3669	6408	11581	14305	13988	10333	7056	5188	6512	9064	7138	4052
Wind	WSAvg	4.1	4.7	4.7	4.1	4.4	5.3	5.2	4.2	3.3	3.0	3.1	2.9	4.5
	PrecAvg	20.6	0.0	0.0	0.1	0.2	1.2	3.0	5.7	6.8	3.1	0.5	0.0	0.0
	PrecMax	29.1	0.0	0.0	0.6	0.7	4.8	5.9	9.4	15.9	6.9	1.4	0.1	0.0
	PrecMin	14.0	0.0	0.0	0.0	0.0	0.0	1.0	2.4	4.1	0.6	0.0	0.0	0.0
	PrecStd	4.3	0.0	0.0	0.1	0.2	1.2	1.3	2.0	3.0	1.7	0.5	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.0	104.2	107.5	109.4	109.0	104.3	99.0	94.5	98.0	102.2	101.8	98.4
		MCWB	63.3	66.9	67.8	71.3	74.7	77.6	78.3	79.0	79.0	71.8	67.6	64.5
	2%	DB	94.7	101.5	105.8	107.9	106.8	101.7	95.9	92.1	95.5	100.5	100.5	95.2
		MCWB	61.2	65.2	67.3	71.1	75.1	77.8	77.6	78.7	79.1	72.7	67.4	62.7
	5%	DB	91.9	98.8	104.3	106.7	104.9	99.6	93.6	90.4	93.7	98.9	99.1	92.8
		MCWB	59.6	63.5	66.8	70.7	75.8	77.7	77.4	78.2	79.0	73.3	66.9	61.5
	10%	DB	88.8	95.8	102.6	105.1	102.9	97.6	91.5	88.3	91.8	97.2	97.3	89.9
		MCWB	58.1	62.4	65.8	70.3	76.3	77.3	77.3	77.8	78.7	74.4	65.8	60.2
	0.4%	WB	64.8	74.4	76.3	78.9	81.5	81.3	80.4	80.6	81.9	81.2	75.2	66.1
		MCDB	94.0	90.5	94.7	101.6	101.1	96.4	92.1	90.3	92.2	93.5	93.0	94.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	62.5	71.6	74.4	77.4	80.1	80.1	79.2	79.7	80.6	79.9	72.5	64.3
		MCDB	92.9	90.1	95.4	100.0	98.2	94.5	89.7	88.6	90.5	92.4	91.9	93.2
	5%	WB	60.3	67.5	71.7	76.3	79.2	79.2	78.7	79.0	79.9	79.0	70.5	62.6
		MCDB	89.6	91.9	96.3	98.3	97.1	93.3	88.6	87.3	89.6	91.4	90.6	90.6
	10%	WB	58.5	63.9	67.5	75.4	78.4	78.7	78.1	78.5	79.2	78.2	68.2	60.9
		MCDB	86.9	94.3	99.5	96.5	95.9	92.6	87.8	86.1	88.7	90.1	90.4	87.5
Mean Daily Temperature Range	5% DB	MDBR	24.7	25.4	26.0	22.7	19.2	16.6	14.7	13.2	16.0	22.2	29.6	25.2
		MCDBR	28.6	28.6	28.5	24.8	21.2	18.7	17.5	15.8	18.6	24.8	31.3	29.1
		MCWBR	11.3	10.4	10.3	8.6	5.8	4.5	4.4	4.8	5.3	6.3	8.7	10.6
	5% WB	MCDBR	28.6	24.7	25.1	20.7	19.0	17.1	15.5	14.5	17.5	20.1	29.0	28.8
		MCWBR	11.8	10.6	12.2	7.6	5.9	5.1	4.8	5.6	5.9	5.8	10.2	11.0
Clear-Sky Solar Irradiance	taub		0.606	0.662	0.739	0.742	0.710	0.671	0.605	0.579	0.576	0.622	0.551	0.541
	taud		1.686	1.539	1.418	1.495	1.553	1.640	1.826	1.876	1.872	1.729	1.773	1.789
	Ebn at Noon		223	217	206	205	207	213	228	238	239	224	235	236
	Edh at Noon		75	91	104	96	89	81	67	65	65	74	69	67
All-Sky Solar Radiation	RadAvg		1747	1883	1981	2033	1981	1897	1817	1720	1873	1918	1825	1719
	RadStd		78	81	103	72	58	65	59	77	67	48	44	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.67	+1.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page